IOP Conference Series:
Materials Science and
Engineering

PURPOSE-LED
PUBLISHING™

PAPER • OPEN ACCESS

Document Tracking and Monitoring of Expenses for State Universities and Colleges Responsibility Units

To cite this article: Joni Neil Capucao *et al* 2020 *IOP Conf. Ser.: Mater. Sci. Eng.* **803** 012042

View the article online for updates and enhancements.

You may also like

- Price-dependent fuzzy demand and shortages scenario with memory effect on inventory system incorporating optimal replenishment strategy
Shilpi Pal, P K Santra and G S Mahapatra
- The Use of Factor Analysis of the Cost When Justifying the Directions of Innovative Development of Crop Production (on the Example of the Grain Complex of the Pskov Region)
I P Voiku, V A Kuchuk and I S Sandu
- The activities and funding of IRPA: an overview
Geoffrey Webb

This content was downloaded from IP address 136.158.254.18 on 23/03/2026 at 06:02

IOP Conf. Series: Materials Science and Engineering **803** (2020) 012042 doi:10.1088/1757-899X/803/1/012042

Document Tracking and Monitoring of Expenses for State Universities and Colleges Responsibility Units

Joni Neil Capucao[1], Sean Sierra[2], Kennedy Cuya[3], Josephine Dela Cruz[4]

[1]College of Arts and Sciences, Partido State University, Camarines Sur, Philippines
[2]Partido State University, San Jose Campus, Camarines Sur, Philippines
[3]College of Arts and Sciences, Partido State University, Camarines Sur, Philippines
[4]University of the Cordilleras, Benguet, Baguio, Philippines

E-mail: [1]jcapucao265@gmail.com, [2]seanasierra519@gmail.com,
[3]kennedy.cuya@parsu.edu.ph, [4]delacruzpen@gmail.com

Abstract. The increasing data volume generated by higher learning institutions, aggregating of expenses, and locating files among tons of others can be a tedious and time-consuming process. Partido State University does not have a document and expense tracking system, making it difficult for the clientele to determine the status of their submitted documents. The development of document tracking and monitoring of expenses system for the University is a responsive, professional committal of the proponents in addressing this gap. The descriptive method explained the existing processes and developmental method in the creation of the system. Semi-structured interview among office personnel revealed the University offices need to adopt online software. In the systems' dry run, the faculty, the staff, and the students gave an overall very good (4.33 and 4.45) ratings for functionality and usability, and also an excellent overall rating of 4.5 for efficiency. The respondents strongly considered the adoption of an online system. The evaluation results manifested by the *Accurateness in Functionality* (4.63, excellent), *Learnability in Usability* (4.63, excellent), and *Time Behaviour in Efficiency* (4.63, excellent) have indicated that the system satisfied all the requirements. After its implementation, if some flaws may surface, adjustments could just be easily adapted.

1. Introduction

The changing technological pressure on management has led to an increasing interest in improving operational processes not merely to enhance performance but as well as staying current. Government organizations in the Philippines, particularly in the academe sector, such as State Universities and Colleges, have experienced various difficulties in implementing efficient processes and process management. These management difficulties are very evident on the following specifics; office transactions are commonly done manually using office applications that frequently lead to a delayed response to inquiries, longer time for dissemination of output, and even reduced service efficiency. If, however, the computerized system has been adopted, they are mostly a standalone application suggesting, therefore, limited access to information. The implementation of online systems, since it has been an intense clamor of interest among practitioners [1], is one of the challenges that need to be addressed. One such justification for this is Republic Act No. 11032, which stressed the need for heightened transparency and guarantees citizen-centric public service and lessen the number of complaints on delayed government transactions with the citizenry allowed to evaluate and give feedback

Content from this work may be used under the terms of the Creative Commons Attribution 3.0 licence. Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.
Published under licence by IOP Publishing Ltd

on public services and the bureaucracy [2]. Also, the ISO 9001:2015 standards further specify the requirements for a quality management system [3] for State Universities and Colleges (SUCs).

Indeed, the large and increasing data volume generated by higher learning institutions, aggregating of expenses, and locating files among tons of others can be a tedious and time-consuming process for administrative and academic staff. Previous works have shown evidence that the challenge can be addressed using an online document tracking system. The system may be designed and developed using responsive website design [4] applicable to both mobile devices and desktop computers [5]. The use of barcodes can even be incorporated into the system for ease of entry [6]. Document Tracking System can be useful in administrative and academic institutions [7] for tracking documents and manage the movement of files from desk to desk and different colleges [8]. A mobile application can be used to track expenses incurred [9].

The absence of online systems in the different responsibility units of Partido State University (PSU) such as the Document Control Office, Finance Office, Colleges, and Campuses creates this difficulty. Such shortcoming makes it difficult for the administrators and personnel in monitoring into how well this office is succeeding in meeting its goals. Since most of the execution of processes are being done manually or mostly using application software such as an electronic spreadsheet, often leads to some specific problems; like difficulty in tracking documents; and monitoring of expenses incurred in the different units; consolidation of outputs; and report generation.

The main goal of the present study is for the proponents to provide a solution to efficiently track the document's status and whereabouts, including other features for aggregating of expenses. In addressing this need, the researchers plan to establish an in-depth validation of data on the information processes that can be designed and adapted for tracking administrative and financial documents in terms of a) flow of documents and b) aggregating of expenses. Next is to develop a system that could help enhance the efficiency in tracking administrative and financial documents through the online system. Finally, it will solicit respondents' (faculty, staff, and students) responses for determining the level of effectiveness on the proposed system. As a whole, the development of document tracking and aggregating of expenses system for the Partido State University is but their professional committal in addressing such research gap and by allowing the University in becoming more responsive.

2. Methodology

The research protocol concerning the design of online document tracking and monitoring of expenses system utilized the descriptive and developmental method. Descriptive method is intended to give focus on the information processes describing the current process flow and the processes that need to be enhanced or replaced in the different responsibility units. For the construction of the online system, the developmental method using rapid application development technology is fitting to fast-track the delivery and implementation of the prototype and for the inclusion of the financial requirements in the next University budget cycle.

2.1. Investigating PSU Existing Processes

Conducting a semi-structured interview with the office personnel and collecting sample forms used by different offices such as work instruction, procedure manual, organizational structure, the master list of logbooks, and observing the office procedures in monitoring document whereabouts and existing status may eventually benchmark in identifying the current procedure in monitoring documents as well as in the aggregation of expenditures in the different responsibility units. These may eventually lead to identifying the data requirements and solutions to the problems in tracking documents and aggregation of expenses. The expected information obtained can be made bases in creating the proposed web-based document tracking and monitoring of expenses system.

2.2. *Development of the Web-based System*

The development of the Document Tracking and Monitoring of Expenses System will focus on handling the various types of documents submitted to the different offices. It is meant to enhance the manner of

locating, storing, and distribution of documents. Likewise, it may expedite the processes in monitoring the expenses incurred by the University's stakeholders.

2.2.1. Web Development Framework. Web Development Framework in this endeavour is the appropriate framework in designing the system. The web development framework is a set of resources and tools for software developers to build and manage web applications, web services, and websites [10]. The Laravel Framework being one of the leading web development frameworks that implement the Model-View-Controller (MVC) context, can be the best choice for a PHP framework that aids developers quickly build secure, upgradable web applications [11]. As shown in figure 1, the Model component pertains to all the data-related logic that the user is working. The View component provides all the User Interface (UI) logic of the application. Moreover, the Controllers act as an interface between Model and View components to process all the business logic and incoming requests, manage data using the Model component, and interact with the Views to render the final output [12].

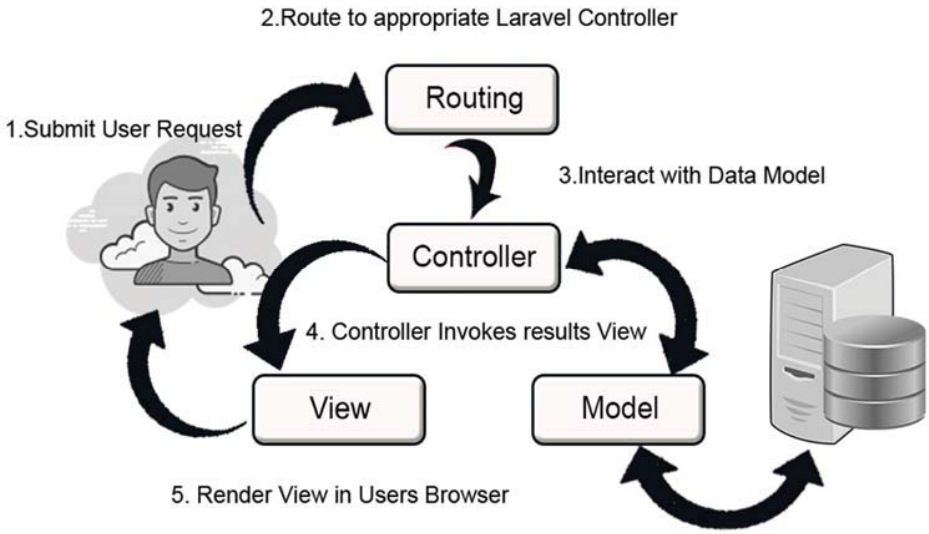


Figure 1. Model, View, and Controller Framework.

2.2.2. Rapid Application Development. For the software development methodology, Rapid Application Development (RAD), shown in figure 2, can be objectively chosen by the researchers, as such it could allow them to get immediate feedback from the end-users to enable necessary adjustments in developing the prototype [13].

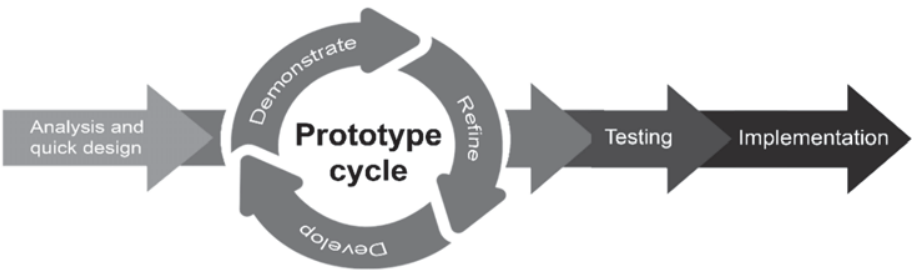


Figure 2. Rapid Application Development (RAD) Methodology.

3. Results and Discussion

3.1. Information Processes Designed and Adapted

This research tried to verify the assumption that information processes designed and Adapted for tracking administrative and financial documents in terms of a) flow of documents and b) aggregation of expenses encountered were essential.

PSU has seven campuses and several administrative and financial offices which, adopts the centralized processing of administrative and financial documents. The primary data source emanates from the Office of the President, Vice-Presidents, Quality Management Office, Document Control Office, Accounting Office, Budget and Finance Office, Cashier Office, College Deans, Campuses, Faculty and Staff, and Students of Partido State University-Main Campus, Goa, Camarines Sur. Additional secondary information collected through grey researches either from the internet, books, related articles, and including theses writings.

It was learned that all of the thirty-two offices of the University were exercising duties and functions as stipulated under the Quality Management Procedures of the ISO 9001:2015 quality management process. These findings affirmed the availability and application of the ISO standards as discovered in the use of Procedure Manuals (PM) and Work Instruction (WI) provided by different auditing offices; General Administration and Support Services (GASS) - 20 offices; Support to Operation (STOs) - 7 offices; and Academics -18 offices. The collected information served basis in determining what documents were processed and the sequence of procedures that were employed. The Master List of Logbooks (508 record books issued) suggested the numerous and complex transactions that transpired from one office to another, although these were all executed through the conventional manual approach, including further the aggregating of financial expenses in the budget and finance office.

3.2. Prototype System Development

The prototype Document Tracking and Monitoring of Expenses System was developed using Rapid Application Development (RAD) methodology and guided by the design of the Use Case Diagram.

3.2.1. Rapid Application Development Methodology. RAD methodology as shown in figure 3 has four phases, namely; 1. Analysis and Quick Design, 2. Prototype Cycle, 3. Testing, and 4. Implementation.

Phase 1: Analysis and quick design. In this stage, the researchers conducted interviews with faculty (n=160), staff (n=77) and students (n=340) of PSU regarding the office flow of documents and the accessibility issues in tracking documents and expense report generation needed in the design of the proposed system.

Phase 2: Prototype cycle. In this phase, the researchers gathered documents such as Procedure Manuals (PM) and Work Instructions (WI) in determining the scope of the system; and identified the needs of the stakeholders in terms of document tracking and aggregating of expenses. These requirements were analyzed, findings of the study were discussed, and the architecture design of the system was made, including the design of the database, components, and modules of the system.

Phase 3: Testing. Under this phase, the researchers analyzed the requirements for the system and designed solutions that were implemented, taking into consideration the requirements, constraints, and all applicable standards and guidelines. The proposed system developed was tested ensuring that it met all the requirements specified.

Phase 4: Implementation. The last part of the RAD methodology was the implementation of the prototype system. The researchers conducted an initial implementation of the prototype delimiting its coverage to those offices with current basic facilities.

3.2.2. Use Case Diagram. The researchers selected the use case modelling, figure 3, as the text-based tool for system analysis and design of the prototype system [14]. The model presents the different roles of the system users. The diagram showed that receiving officers in the university are responsible in encoding details of the documents, generating barcode and issuance of electronic mail to the clientele.

Head of offices can access the documents, check document entry and recommend approval. The president has provisions for checking the user performance logs and approval of documents. The clients are not allowed to access the important features of the system and can only check and track the status of their submitted documents and view other system information.

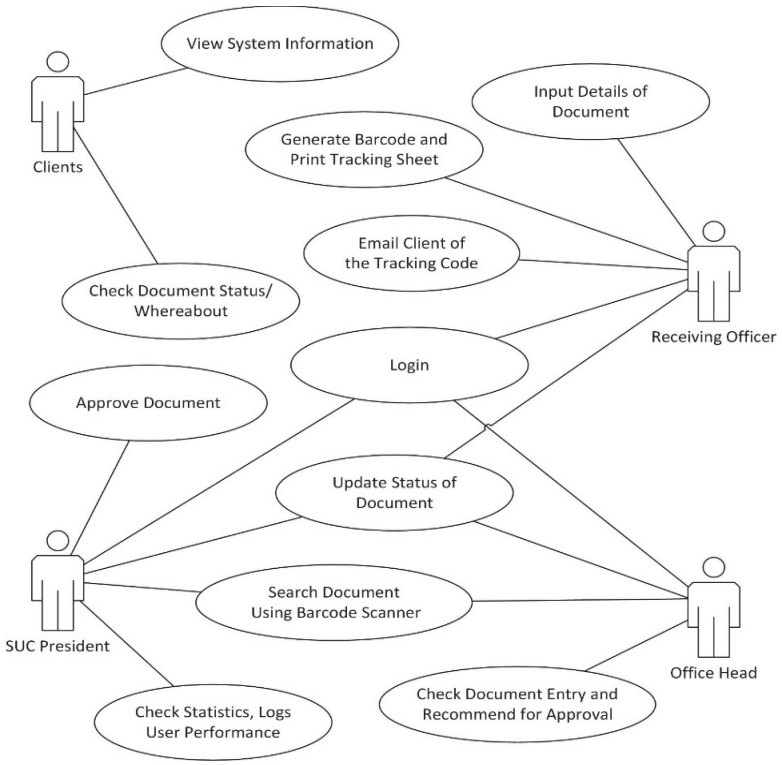


Figure 3. Use Case Diagram.

3.2.3. User Interface. Figure 4 shows sample users' interfaces, indicating among other things that this prototype system can be accessed either through mobile phones or laptop/desktop computers by respective clients and other interested parties.

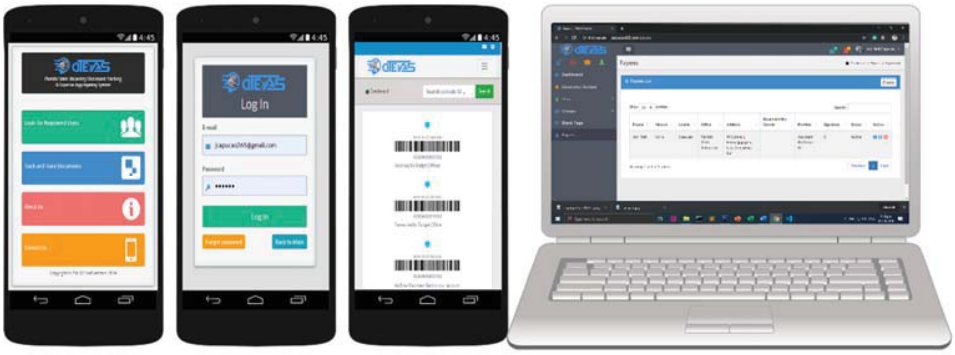


Figure 4. User Interface for Mobile Phone and Desktop Computer.

3.3. Evaluation of the Proposed System

In assessing the prototype systems' functionality, usability, and efficiency the researchers collected primary data from the total sample size of five hundred seventy-seven (577) represented by faculty members (160), staff (77), and students (340) of Partido State University. This number was within the five percent tolerable margin of sampling error, though purposively (homogenous) selected the intent is to provide higher representative (see table 1).

Table 1. Distribution of Respondents.

Faculty		Staff		Students	
Target Population	Sample Size	Target Population	Sample Size	Target Population	Sample Size
271	160	96	77	2500	340

The evaluation result of the proposed system in terms of: (a) functionality; (b) usability; and (c) efficiency using the ISO 9126 Quality model for External and Internal Quality is shown in table 2. All responses along functionality (4.33, very good), usability (4.45, very good) and efficiency (4.50, excellent) showed no significant difference at alpha .05, suggesting therefore that faculty, staff and students have the same observations favorably indorsing the use of the said prototype system. There were observable low scores in Interoperability as rated by faculty (3.90) and staff (3.90). The respondents manifested that the proposed system provides an avenue to input necessary information about the whereabouts of the documents, expenses incurred, and presents reports generated by the system but is not meant to exchange information with other systems adequately. For Understandability, being new software, the expected interfaces to quickly understand the flow of the system suggests getting more used to it (faculty, 4.10; staff, 4.10). Under Security, the online programming language chosen is one of the general types. Hence, faculty (4.2) and staff (4.00), provided leeway for enhancement.

The adoption of the system may enhance the tracking of documents' whereabouts, and aggregation of expenses with the conversion of unscientific manual processes to online system shows the potential of meeting the main objective of the study as reflected by the excellent (4.50) rating in Efficiency. Respondents' overall rating of very good (4.33, 4.45) for functionality and usability suggests that the prototype meets the expected requirements. Thus, the implementation of such a system by State Universities and Colleges Responsibility Units, under the Philippine context, answers their utmost administrative and finance needs.

Table 2. ISO 9126 Quality model for External and Internal Quality.

Characteristic	Sub-characteristic	Explanation	Weighted Mean				Interpretation
			Faculty	Staff	Students	Total	
Functionality	Suitability	Can software perform the tasks required?	4.30	4.40	4.60	4.43	Very Good
	Accurateness	Is the result as expected?	4.50	4.50	4.90	4.63	Excellent
	Interoperability	Can the system interact with another system?	3.90	3.90	4.30	4.03	Very Good
	Security	Does the software prevent unauthorized access?	4.20	4.00	4.50	4.23	Very Good
Usability	Understandability	Does the user comprehend how to use the system easily?	4.10	4.10	4.30	4.17	Very Good
	Learnability	Can the user learn to use the system easily?	4.50	4.50	4.90	4.63	Excellent
	Operability	Can the user use the system without much effort?	4.50	4.40	4.80	4.57	Excellent
	Attractiveness	Does the interface look good?	4.30	4.40	4.60	4.43	Very Good
Efficiency	Time Behaviour	How quickly does the system respond?	4.60	4.50	4.80	4.63	Excellent
	Resource Utilization	Does the system utilize resources efficiently?	4.30	4.30	4.50	4.37	Very Good

H_o *Significant @.05 **Significant @.01

4. Conclusion

In this study, a prototype document tracking and monitoring of expenses system were presented using an MVC web development framework. The PSU responsibility unit's business processes considered and exhibited using a use case diagram. In the systems' dry run, the faculty, the staff, and the students gave an overall very good (4.33 and 4.45) ratings for functionality and usability, and also an excellent overall rating of 4.5 for efficiency. The respondents strongly recommend the adoption of the online system. The evaluation results manifested by the *Accurateness in Functionality* (4.63, excellent), *Learnability in Usability* (4.63, excellent), and *Time Behaviour in Efficiency* (4.63, excellent) have indicated that the system satisfied all the requirements. After its full implementation, if some minor flaws may surface, adjustments could just be easily adapted.

References

[1] Rhee S, Cho N and Bae H 2008 Increasing the efficiency of business processes using a theory of constraints *Information Systems Frontiers* **12** 4 pp 443–55
[2] GovPh 2018 The implementing rules and regulations of RA.11032 *Official Gazette* Available from: https://www.lawphil.net/statutes/repacts/ ra2018/pdf/ra_11032_2018.pdf
[3] ISO/TC176/SC2 ISO9001:2015 2015 *Quality management systems requirements* Available from: https://www.iso.org/standard/62085.html
[4] Giurgiu L and Gligorea I 2017 Responsive web design techniques *Int. Conf. Knowledge-Based Organization* **23** 10.1515/kbo-2017-0153
[5] Waseem B I and Abdelaziz H I 2016 Responsive web design techniques *Int. J. of Comp. App.* **150** 2 pp 18-27
[6] Gao J Z, Prakash L and Jagatesan R 2007 Understanding 2D-barbode technology and applications in m-commerce design and implementation of a 2D barcode processing solution *31st Annual Int. Computer Software and Applicatioin Conf.*
[7] Ajoge N S and Hamisu I A 2015 Document e-tracker: a panacea for efficient internal document management system in kaduna polytechnic *JOSTE* **3** 2 pp 169-81
[8] Omoregbe N A, Azeta A A and Adewumi A 2014 Development of a file tracking system for tertiary institutions *Proc. of edulearn14 Conf.* pp 3963-70
[9] Thanapal M P, Patel Y, Lokesh R T P and Satheesh K J 2015 *Income and expense tracker* (Tamil Nadu, India: Vellore)
[10] Ryabtsev A *Web frameworks: how to get started* Available from: https://djangostars.com/blog/ what-is-a-web-framework
[11] Armel J *Web application development with laravel PHP framework version 4* Available from: https://www.theseus.fi/bitstream/handle/10024/74052/Author.pdf
[12] Majeed A and Rauf I 2018 MVC architecture: a detailed insight to the modern web applications development *J. of Solar & Photoenergy Systems* **1** pp 1-7
[13] Kralev V S and Kraleva R S 2017 Methods and tools for rapid application development *Proc. of the III Int Scientific and Practical Conf.*:*Methodology of Modern Research"* pp 21-4
[14] Gemino A and Parker D 2009 Use case diagrams in support of use case modeling: deriving understanding from the picture *J. of Database Management* **20** (1)